

ESTIMATING GALAXY LUMINOSITY FUNCTIONS

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ABSTRACT

In this work a comparison between different galaxy luminosity function estimators by means of Monte Carlo simulations is presented. The simulations show that the C^- method of Lynden-Bell (1971, MNRAS, 155, 95) and the STY method derived by Sandage, Tammann & Yahil (1979, ApJ, 232, 352) are the best estimators to measure the shape of the luminosity function. The simulations also show that the STY estimator has a bias such that the faint-end slope is underestimated for steeper inclinations of the Schechter Function, and that this bias becomes quite severe when the sample contains only a few hundred objects. Overall, the C^- is the most robust estimator, being less affected by different values of the faint end slope of the Schechter parameterization and sample size. The simulations are also used to compare different estimators of the luminosity function normalization. They demonstrate that most methods bias the recovered mean density towards values which are $\sim 20\%$ lower than the input value. © 1997 American Astronomical Society. [S0004-6256(97)03909-5]

1. INTRODUCTION

The luminosity function of galaxies is an important tool in the study of large-scale structure, as it ultimately allows the estimation of the total content of luminous matter in the form of galaxies (for a comprehensive review on many applications of the luminosity function the reader is referred to Binggeli *et al.* 1988). As described by Binggeli *et al.* (1988), over the years several methods have been proposed to calculate the luminosity function and have been applied to a variety of samples of field (and cluster) galaxies, as well as quasars. The first determinations of the luminosity function simply counted the number of sample objects inside a given volume $\Phi = N/V$, and were used in early works (e.g., Hubble 1936), but, as noted by Binggeli *et al.* (1988), detailed descriptions of the method only appeared much later (Trumpler & Weaver 1953; Christensen 1975; Schechter 1976). This has come to be known as the *classical method*, a term coined by Felten (1976), and is based on the assumption that the distribution of galaxies in space is uniform (Binggeli *et al.* 1988). An estimator derived from the classical method was published by Schmidt (1968) to study quasar evolution, which takes into account a weight inversely proportional to the luminosity of the object. This is known as the $1/V_{\max}$ method, but as in the case of the classical estimator, it contains the underlying assumption that the distribution is uniform. This estimator was first applied to a sample of galaxies by Huchra & Sargent (1973), while a paper describing how to combine different samples coherently was presented by Avni & Bahcall (1980). A further development using the $1/V_{\max}$ method was presented by Eales (1993) who used it to calculate the luminosity function as a function of redshift.

In order to overcome the dependence on the assumption of a uniform distribution, Lynden-Bell (1971) derived the C^- method, which was also applied to the sample of quasars

studied by Schmidt (1968). Lynden-Bell (1971) showed that the C^- estimator is a maximum-likelihood method, and requires no assumption on the distribution, save that the luminosity function is of the same shape at all points along the line of sight, and that the sample should be ordered in luminosity. Further work on the C^- was carried out by Jackson (1974) who extended it so as to analyze several samples, where the selection function of each sample is taken into account. By assuming that the distribution of objects could be described by analytic functions, Jackson (1974) also obtained error estimates for this method. A method similar to the C^- was proposed by Nicoll & Segal (1983), but differs in the sense that it uses binning, in contrast to Lynden-Bell (1971), where no binning is used. The application of the C^- method to galaxies was first proposed by Choloniewski (1987), who by using simple arguments showed that it can give a properly normalized luminosity function, without making any assumption on the shape of the density or luminosity distributions.

A method developed to remove effects caused by the density inhomogeneities was derived by Turner (1979) and independently by Kirshner *et al.* (1979), who considered the ratio of the differential luminosity function at each absolute magnitude interval between M and $M + dM$ and the total number of galaxies brighter than M . A slightly modified version of this method binning galaxies in equal radial velocity intervals instead of equal magnitude bins was used by Davis & Huchra (1982, hereafter DH) and de Lapparent *et al.* (1989, hereafter LGH).

Another maximum-likelihood estimator was developed by Sandage *et al.* (1979, hereafter STY), who applied it to the Revised Shapley Ames catalog of galaxies (Sandage & Tammann 1981). This estimator also cancels out the contribution of the density distribution, and allows including corrections for galaxy incompleteness or other effects. In contrast to the

C^- , this method assumes that the luminosity distribution is described by an analytic function. In order to provide a visual representation as well as an estimate of the goodness-of-fit of the STY maximum-likelihood, Efstathiou *et al.* (1988, hereafter EEP) derived the stepwise maximum likelihood method (SWML) where galaxies are counted in magnitude bins, but where no functional shape is assumed. The stepwise maximum likelihood has been extended by Heyl *et al.* (1997) to take into account the dependence of the luminosity function with redshift, and by Springel & White (1997) who instead of assuming the non-parametric luminosity function as being described by a constant value in each magnitude bin (as in a histogram), assume power-laws so that the distribution becomes smoother.

Two further maximum-likelihood and clustering insensitive methods were derived by Marshall *et al.* (1983) and Choloniewski (1986) by using as a basic assumption the fact that the distribution of objects is the result of a random process described by a Poisson probability distribution (Binggeli *et al.* 1988). Marshall *et al.* (1983) assumed that the luminosity and density distributions could be described by an analytic form, and applied it to a sample of quasars. On the other hand, Choloniewski's (1986) derivation is non-parametric, but requires binning the data, and was applied by its inventor to a sample of galaxies taken from the CfA1 survey (Huchra *et al.* 1983).

In spite of the variety of methods that have been proposed to calculate the luminosity function (Binggeli *et al.* 1988 list 13 in their table 2), no detailed comparison between them has been carried out. For instance, although many possibilities have been forwarded to explain the observed discrepancy between the normalizations measured for the surveys of galaxies nearby and at intermediate redshifts ($z \leq 0.1$) (e.g., da Costa *et al.* 1994; Marzke *et al.* 1994, hereafter MHG; Lin *et al.* 1996) relative to distant samples (e.g., Lilly *et al.* 1995; Ellis *et al.* 1996), it is not clear whether this could not be partly due to the procedures used in the calculation of the luminosity function. It is interesting to note that the nearby surveys have relied on the combination of the STY maximum-likelihood with the SWML, while the surveys of distant galaxies have used primarily the $1/V_{\max}$ estimator. The results from these works show that in general the deeper samples present higher values of the normalization than local ones and could imply in the existence either of density evolution or a population of "disappearing" galaxies. Other explanations, such as a poor determination of the faint end of the local luminosity function (Gronwall & Koo 1995; Koo, *et al.* 1993), have also been proposed and are supported by recent determinations made by da Costa *et al.* (1997) who find a faint end slope $\alpha \sim -1.2$ for the SSRS2, while an even steeper slope $\alpha \sim -1.5$ has been determined by Sprayberry *et al.* (1997) using a survey of low surface brightness galaxies.

The main motivation of this paper is to examine how well some of the different estimators can measure the luminosity function in an apparent magnitude-limited survey. This will be done by comparing the results obtained applying these techniques to simulated catalogs of galaxies, and will examine if the use of different estimators could be a possible

source of the discrepancies obtained by the different redshift surveys. Comparisons between some of the methods have appeared previously (Choloniewski 1986; EEP; LGH; Ratcliffe *et al.* 1997), while discussions on the properties of some estimators have been presented by Petrosian (1992) and Heyl *et al.* (1997).

This paper is organized as follows: Section 2 will describe the estimators; in Section 3 we describe the Monte Carlo simulations and the results we obtain. The application of the different estimators to a sample of real galaxies is presented in Section 4, followed by the conclusions in Section 5.

2. ESTIMATORS

2.1 STY Estimator

The first estimator we will describe is the parametric maximum-likelihood estimator proposed by STY. This estimator is insensitive to the presence of fluctuations in the sample (e.g., STY; MHG), and does not require binning the data, so that all the information contained in the sample is preserved. In the STY formalism, one defines the cumulative probability that a galaxy at redshift z will have a magnitude brighter than M as

$$P(M, z) = \frac{\int_{-\infty}^M \phi(M') \rho(z) f(m') dM'}{\int_{-\infty}^{+\infty} \phi(M') \rho(z) f(m') dM'} \quad (1)$$

where $\phi(M)$ is the differential luminosity function, $\rho(z)$ represents the redshift distribution at z and $f(m')$ the catalog incompleteness, where $m' = M' + 5 \log[D_L(z)] + 25 + K(z, T)$ is the apparent magnitude; $D_L(z)$ is the luminosity distance and $K(z, T)$ the cosmological K -correction which depends on the redshift z and morphological type T . In this notation, one may easily incorporate other effects such as sampling rates (Lin *et al.* 1996), and the magnitude error (EEP; MHG). The probability density that a galaxy will be detected in a redshift survey is obtained by calculating the partial derivative of $P(M, z)$ relative to M :

$$p(M_i, z_i) = \frac{\phi(M_i)}{\int_{-\infty}^{M_{\text{faint}}(z_i)} \phi(M) dM}. \quad (2)$$

Thus this probability is directly proportional to the value of the differential luminosity function at M_i and inversely proportional to the faintest absolute magnitude visible at z_i , $M_{\text{faint}}(z_i)$ (Heyl *et al.* 1997). One can then define the likelihood which is the joint probability of all galaxies in the sample belonging to the same parent distribution:

$$\mathcal{L} = \prod_{i=1}^{N_g} p(M_i, z_i). \quad (3)$$

To estimate the maximum likelihood, one must maximize this function relative to the $p(M_i, z_i)$. To do this, one has to assume some parameterization of the luminosity function, such as a Schechter (1976) function, and maximizing the likelihood relative to the parameters M^* and α . The errors can be estimated from the error ellipsoid defined as

$$\ln \mathcal{L} = \ln \mathcal{L}_{\max} - \frac{1}{2} \chi^2_{\beta}(N) \quad (4)$$

where $\chi^2_{\beta}(N)$ is the β point of the χ^2 distribution with N degrees of freedom (EEP). Because this method involves ratios between the differential and integrated luminosity functions, and we are assuming that the fluctuations are independent of $\phi(M)$, the normalization has to be calculated independently.

2.2 SWML

As discussed by EEP, the STY maximum likelihood presents the inconvenience that one cannot test whether the parameterization represents a good fit to the data. In order to provide an adequate visual representation of the data, EEP derived the Stepwise Maximum Likelihood method, which does not rely on the assumption of a simple functional form for $\phi(M)$. This is done by parameterizing $\phi(M)$ as a series of N_p steps:

$$\phi(M) = \phi_k, \quad M_k - \Delta M/2 < M < M_k + \Delta M/2, \quad k = 1, \dots, N_p. \quad (5)$$

Let $x = M_i - M_k$. We can then define two window functions (EEP; MHG):

$$W(x) = \begin{cases} 1, & |x| \leq \Delta M/2, \\ 0, & |x| > \Delta M/2, \end{cases} \quad (6)$$

so that $W(x)\phi_k$ is the differential luminosity function at M_k , and

$$H(x) = \begin{cases} 1, & x \leq -\Delta M/2, \\ \frac{1}{2} - \frac{x}{\Delta M}, & -\Delta M/2 \leq x \leq +\Delta M/2, \\ 0, & x \geq +\Delta M/2. \end{cases} \quad (7)$$

One may then re-write the denominator in Eq. (2) as:

$$\sum_{i=1}^{N_g} \left\{ \sum_{j=1}^{N_p} \phi_j \Delta M H[M_j - M_{\text{faint}(z_i)}] \right\}. \quad (8)$$

The likelihood function can then be written as

$$\ln \mathcal{L} = \sum_{i=1}^{N_g} W(M_i - M_k) \ln \phi_k - \sum_{i=1}^{N_g} \left\{ \sum_{j=1}^{N_p} \phi_j \Delta M H[M_j - M_{\text{faint}(z_i)}] \right\} + \text{constant}. \quad (9)$$

The likelihood is calculated using an arbitrary normalization, and in order to compare different surveys, EEP imposed an additional constraint:

$$g = \sum_{k=1}^{N_p} \phi_k \Delta M 10^{0.4\beta(M_k - M_f)} - 1 = 0 \quad (10)$$

where β is a constant, chosen by EEP to be ~ -1.5 in order to allow minimum variance, and M_f is a fiducial magnitude. By including this constant with a Lagrangian multiplier, one defines

$$\ln \mathcal{L}' = \ln \mathcal{L} + \lambda g(\phi_k), \quad (11)$$

which is maximized relative to the ϕ_k and λ using condition (10) and $\lambda = 0$. The ϕ_k are obtained iteratively from

$$\phi_k \Delta M$$

$$= \frac{\sum_{i=1}^{N_g} W(M_i - M_k)}{\sum_{i=1}^{N_g} \left\{ H[M_k - M_{\text{faint}(z_i)}] / \sum_{j=1}^{N_p} \phi_j \Delta M H[M_j - M_{\text{faint}(z_i)}] \right\}}. \quad (12)$$

This method also allows incorporating effects such as the catalog incompleteness (e.g., Marzke & da Costa 1997). Because of the arbitrary normalization, the mean density must be estimated independently. The errors for the ϕ_k may be estimated from the covariance matrix, which is the inverse of the information matrix (EEP):

$$\text{cov}(\phi_k) = [\mathbf{I}(\phi_k)]^{-1} = \begin{bmatrix} \frac{\partial^2 \ln \mathcal{L}}{\partial \phi_i \partial \phi_j} + \frac{\partial g}{\partial \phi_i} \frac{\partial g}{\partial \phi_j} & \frac{\partial g}{\partial \phi_i} \\ \frac{\partial g}{\partial \phi_i} & 0 \end{bmatrix}^{-1}_{\phi = \phi_k} \quad (13)$$

2.3 Choloniewski Method

A similar method to the SWML was proposed by Choloniewski (1986). In this method galaxies are binned in equal intervals of absolute magnitudes and distance modulus (μ). By assuming that the distribution of galaxies in the $M - \mu$ plane is described by a Poisson process, the probability of finding N_{ij} galaxies in a cell of width dM is

$$P(N_{ij}) = \exp(-\lambda_{ij}) \frac{\lambda_{ij}^{N_{ij}}}{N_{ij}!} \quad (14)$$

where

$$\lambda_{ij} = \frac{1}{n} \phi_i \Delta M \rho_j \frac{\Omega}{3} (r_j^3 - r_{j-1}^3); \quad (15)$$

$\bar{n}$ is the mean density of the sample, ϕ_i the mean value of the luminosity function in the ΔM interval and ρ_j the local value of the galaxy density at a distance r_j , given by the distance modulus μ_j . The likelihood for a magnitude-limited sample is then

$$\mathcal{L} = \prod_{i=1}^A \prod_{j=1}^B \exp(-\lambda_{ij}) \frac{\lambda_{ij}^{N_{ij}}}{N_{ij}!} \quad (16)$$

where A and B are the number of bins in magnitudes and distance modulus, respectively, and S represents the selection line given by $S = M + \mu$, μ being the distance modulus of an object with absolute magnitude M when it is at the survey's limiting apparent magnitude m_{lim} . This expression is formally identical to Eq. (3) in the STY method, as we are also considering a product of probabilities. As described by Choloniewski (1986), this likelihood is maximized relative to the parameters $E_k = (\bar{n}, \phi_i, \rho_j)$, and a non-parametric estimate is obtained both for the luminosity function and the density distribution. However, as pointed out by Choloniewski (1986), this estimator does not use all information available in the sample, even for objects which are contained

in the survey. The reason for this is that when one projects the distribution of galaxies on the $M-\mu$ plane, only those cells which are fully contained within the survey limits are used. Cells which are at the survey limit and thus only partially contained are ignored. One could in principle solve this by using bins as small as possible, but, as discussed by Choloniewski (1986), smaller bins tend to give more biased results and larger errors in the determination of the likelihood. The maximum-likelihood is estimated iteratively and converges rather fast. Once the maximum-likelihood parameters are known, one may fit the non-parametric ϕ_k distribution with a parametric representation. The uncertainties for the individual estimators may be calculated in a similar fashion as in the SWML method, by means of the covariance matrix:

$$\text{cov}(E_k) \approx \left[\frac{\partial^2 \mathcal{L}}{\partial E_i \partial E_j} \right]_{E=E_k}^{-1}. \quad (17)$$

This method differs from the STY and SWML in the sense that the mean density, and therefore the normalization of the sample, is estimated simultaneously with the luminosity function.

2.4 Turner Method

The Turner (1979) method uses the same assumptions as the STY estimator, but differs in the way the luminosity function is calculated. As in the STY method, for an absolute magnitude M' , the expected number of objects in a survey is given by

$$\langle n(M') dM \rangle = \rho(M') \phi(M') dM \quad (18)$$

where $\rho(M')$ is the density distribution. The number of objects with absolute magnitudes brighter than M' and apparent magnitudes brighter than the survey limit is

$$\langle N(M') dM \rangle = \rho(M') \int_{-\infty}^{M'} \phi(M) dM. \quad (19)$$

One then defines a quantity, where the density cancels out:

$$Y(M') dM = \frac{\langle n(M') dM \rangle}{\langle N(M') dM \rangle} = \frac{\phi(M') dM}{\int_{-\infty}^{M'} \phi(M) dM} = \frac{d\Psi}{\Psi}, \quad (20)$$

where Ψ is the integral luminosity function. This expression is identical to equation (2). One can make the approximation

$$Y(M') \approx \frac{d \ln \Psi(M \leq M')}{dM}. \quad (21)$$

The integration of this equation yields the integrated luminosity function and the differential luminosity function can be derived by re-writing equation (20) as $\phi(M) dM = \Psi(M) Y(M) dM$ (LGH) or:

$$\phi(M) \approx \phi_0 Y(M) \exp \left\{ \int_{-\infty}^M Y(M) dM \right\} \quad (22)$$

where ϕ_0 is an integration constant.

In practice, one calculates $Y(M)$ in magnitude bins (LGH)

$$Y(M) \delta M = \frac{N(\leq M) - N(\leq M - \delta M)}{N(\leq M)} \quad (23)$$

which we represent as δM , as they may be linear in magnitudes (Turner 1979; Kirshner *et al.* 1979), or a function derived from linear bins in redshift (DH; LGH). The errors associated to $Y(M)$ may be assumed as following a Poisson distribution (e.g., LGH).

2.5 C^- Method

The C^- method was derived by Lynden-Bell (1971) and was extensively analyzed by Jackson (1974), Choloniewski (1987) and Subbarao *et al.* (1996). The latter also adapted this method so it could be applied to a sample with photometric redshifts. The C^- method is the limiting case of the last three methods described above, when each bin contains one only object. With the C^- method one recovers the integral luminosity function $\Psi(M_i)$, and the differential luminosity function may be obtained through $\phi(M) = -d\Psi/dM$. Following Lynden-Bell (1971) we call the observed sample at M of $X(M)$. In general, for a change dM in the magnitudes, the change in the number of observed objects is not the same as the change in the entire distribution (Subbarao *et al.* 1996), because galaxies fainter than the apparent magnitude limit are undetected:

$$\frac{d\Psi}{\Psi} > \frac{dX}{X}. \quad (24)$$

However, one may construct $C(M)$, a subset of $X(M)$ where one makes the assumption that for an infinitesimal increase dM the variation in the observed distribution should be equal to the change in the overall distribution:

$$\frac{dX}{C} = \frac{d\Psi}{\Psi}. \quad (25)$$

The function $C(M)$ represents the total number of observed objects with absolute magnitudes brighter than M . Expression (25) can be integrated to yield the integrated luminosity function

$$\begin{aligned} \Psi(M) &= A \exp \left\{ \int_{-\infty}^M \frac{dX}{C} \right\} \\ &= A \exp \left\{ \int_{-\infty}^M \frac{1}{C(M)} \frac{dX(M)}{dM} dM \right\}, \end{aligned} \quad (26)$$

which is similar to the expression obtained in the Turner method.

In practice, for a redshift survey one may write $[dX(M)]/dM$ as a series of Dirac delta functions. However, because at the points $M = M_i$, $C(M)$ is indeterminate (Lynden-Bell 1971; Subbarao *et al.* 1996), one must consider the following approximation (Lynden-Bell 1971) for each M_i :

$$C(M) = C^-(M_i) + x \quad (27)$$

where $x = X(M^+) - X(M)$ is the variation of the number of galaxies in the interval $M^+ - M$ and $C^-(M_i)$ is the total number of objects with magnitudes brighter than M_i , but

with the point i itself omitted (Lynden-Bell 1971). One may then integrate over the interval just around M_i :

$$\int_{M_{i-}}^{M_{i+}} \frac{1}{C(M_i)} dX(M_i) = \int_0^1 \frac{dx}{C^- + x} = \ln \left[\frac{C^-(M_i) + 1}{C^-(M_i)} \right]. \quad (28)$$

One may then rewrite (26) as

$$\Psi(M_k) = A \exp \sum_{j=1}^k \ln \left[\frac{C^-(M_j) + 1}{C^-(M_j)} \right] = A \prod_{j=1}^k \frac{C^-(M_j) + 1}{C^-(M_j)}. \quad (29)$$

With this expression one may construct the non-parametric luminosity function, while the integration constant A will represent the normalization of the luminosity function. The normalization constant can be obtained from any non-zero number, and working up and down the magnitude range using factors of $[C^-(M_i) + 1]/C^-(M_i)$ and its inverse (e.g., Lynden-Bell 1971). Jackson (1974) shows how the normalization can be obtained when several samples are combined. This is done by determining the expected number of objects in each subsample using an expression analogous to (39) below, where each subsample is weighted by the appropriate selection function. The normalization is then obtained comparing the expected number with total number of observed objects. Jackson (1974) also showed how to estimate errors for the C^- method, by assuming that the distribution may be described analytically and then calculating the covariance matrix, in a completely analogous way as the SWML and Choloniewski methods.

2.6 $1/V_{\max}$

The last method that will be described is the $1/V_{\max}$ method which was first published by Schmidt (1968). A detailed study of this method was carried out by Felten (1976), who showed it is a minimum-variance maximum-likelihood estimator. The method assumes that for a given absolute magnitude

$$\Psi(M) = \sum_{j=1}^{N_g} \frac{1}{V_{\max}(j)}, \quad (30)$$

where $V_{\max}(j)$ is the volume corresponding to the maximum distance galaxy j could be observed, and still be included in the sample. One may also calculate the differential luminosity function using a variant of the “classical” estimator, by counting galaxies in magnitude bins as done by DH:

$$\phi(M) dM = \frac{\sum_{i=1}^{N_g} N(M - dM/2 \leq M_i \leq M + dM/2)}{V_{\max}(M)}. \quad (31)$$

Felten (1976) demonstrated that even though this estimator is biased, because one loses information on where about in the magnitude bin a galaxy is located, for small enough intervals dM , it does provide a reasonable estimate of the luminosity function. The $1/V_{\max}$ estimator has one advantage over some of methods described above, in the sense that it gives simultaneously the shape and the normalization of the luminosity function. However, this is also its major drawback, because it is very sensitive to density fluctuations.

3. NORMALIZATION OF THE LUMINOSITY FUNCTION

Most of the methods described in the previous section recover the shape of the luminosity function, but, with the exception of the $1/V_{\max}$ and the Choloniewski estimators (and in special cases the C^-) they do not provide any information about the normalization (ϕ^*), which has to be estimated in an independent manner. This quantity is related to the mean density ($\bar{n}$) of the sample through:

$$\phi^* = \frac{\bar{n}}{\int_{M_{\text{bright}}}^{M_{\text{faint}}} \phi(M) dM} \quad (32)$$

where M_{bright} and M_{faint} are the brightest and faintest absolute magnitudes considered in the survey. The mean density is obtained correcting the redshift distribution by the selection function, which describes the probability of a galaxy at redshift z being included in a survey (DH; EEP; LGH):

$$s(z) = \frac{\int_{M_{\text{bright}}}^{\min(M_{\max}(z_i), M_{\text{faint}})} \phi(M) dM}{\int_{M_{\text{bright}}}^{M_{\text{faint}}} \phi(M) dM}. \quad (33)$$

where $M_{\max}(z_i)$ is the faintest absolute magnitude detected at redshift z_i . Various methods have been proposed to calculate the mean density. DH derived a minimum-variance estimator:

$$n = \left[\sum_{i=1}^{N_g} N_i(z_i) w(z_i) \right] / \left[\int_0^{z_{\max}} s(z) w(z) \frac{dV}{dz} dz \right] \quad (34)$$

where $N_i(z_i)$ is the number of galaxies at redshift z_i and the weights are inversely proportional to the selection function and the second moment of the two-point correlation function, which represents the mean number of galaxies in excess of random around each galaxy out to a distance r :

$$w(z_i) = \frac{1}{1 + \bar{n} J_3 s(z)}, \quad J_3 = \int_0^r r^2 \xi(r) dr. \quad (35)$$

In this case, the density is obtained through iteration, and can be calculated either by counting galaxies in redshift bins (e.g., MHG) or by considering bins containing only 1 galaxy. If one sets $w(z_i) = 1$, expression (34) reduces to the n_3 estimator of DH:

$$n_3 = \frac{N_T}{\int_0^{z_{\max}} s(z) dz}. \quad (36)$$

This estimator is fairly robust, but is also affected by large-scale inhomogeneities in the foreground (DH). Another estimator proposed by DH is

$$n_1 = \left[\int_0^{z_{\max}} \frac{N(z)}{s(z)} dz \right] / \left[\int_0^{z_{\max}} \frac{dV}{dz} dz \right]. \quad (37)$$

$N(z)$ in this equation represents the number of galaxies luminous enough to be included in a shell of width dz at redshift z . This procedure gives higher weights to the more distant points where the selection function becomes less certain. DH solved this potential bias by limiting the determination to the range where $s(z) \geq 0.1$, while an alternative to

truncation at a given value of $s(z)$ was made by LGH who calculated the derived ϕ^* from the median and mean values of the n_1 and n_3 estimators calculated over the entire depth of the survey.

A slightly different estimator from the above was proposed by EEP, where the contribution of the two-point correlation function is not taken into account, and is the n_1 estimator at the limiting case of only one galaxy per bin:

$$n_{\text{eep}} = \frac{1}{V} \sum_{i=1}^{N_g} \frac{1}{s(z_i)}. \quad (38)$$

Finally, another estimator proposed by EEP is the normalization by means of the observed number counts. To obtain the normalization, one calculates the expected number of galaxies by integrating the luminosity function over the whole surveyed volume for each apparent magnitude bin:

$$A(M) = \int_0^\infty dz \frac{dV}{dz} \int_{-\infty}^{M_{\text{max}}(z)} \phi(M) dM = \phi^* I(m) \quad (39)$$

and the value of ϕ^* is obtained by minimizing

$$\sum_n \frac{[dN(m_n) - \phi^* dI(m_n)]^2}{\phi^* dI(m_n)} \quad (40)$$

where $dN(m_n)$ is the observed number of galaxies to the limiting apparent magnitude m_n . Following EEP, the normalization is then:

$$\phi_c^* = \left[\frac{\sum_n [dN(m_n)]^2 / dI(m_n)}{\sum_n dI(m_n)} \right]^{1/2}. \quad (41)$$

The uncertainties in the mean density have only been derived in the case of the minimum-variance estimator (DH):

$$\frac{\delta \bar{n}}{\bar{n}} = \left[\frac{1}{\int_0^{z_{\text{max}}} s(z) w(z) (dV/dz) dz} \right]^{1/2} \sim \left[\frac{J_3}{V} \right]^{1/2} \quad (42)$$

and when calculating the uncertainty of ϕ_{minvar} one should also take into account the uncertainties in α and M^* (Lin *et al.* 1996). Alternatively, to estimate the error in the density one could use the dispersion around the mean or median values of the density counted in shells. By using this procedure, LGH estimated that the uncertainty in the determination of ϕ^* is $\sim 25\%$ for the CfA2 slices.

4. THE MONTE CARLO SIMULATIONS

For the Monte Carlo simulations the transformation method (Press *et al.* 1986) was used to generate random variates first in redshift and then in magnitude. A homogeneous redshift distribution is assumed, and is obtained by calculating for a given interval in redshift

$$N(z) = \int_{z_1}^{z_2} \int_{-\infty}^{M_{\text{max}}(z')} \phi(M) \frac{dV}{dz'} dM dz', \quad (43)$$

where dV/dz' is the volume element corrected for relativistic effects. In this work we use the $\phi(M)$ as parameterized by Schechter (1976), which expressed in magnitudes is

$$\phi(M) dM = 0.4 \ln 10 \phi^* 10^{0.4(M^* - M)(\alpha + 1)} \times \exp\{-10^{0.4(M^* - M)(\alpha + 1)}\} dM \quad (44)$$

where ϕ^* is the normalization, M^* a fiducial magnitude that characterizes the point where the growth of the function changes from a power-law to an exponential slope, the inclination of which is described by α . Although other parameterizations for the luminosity function have been proposed (e.g., Yahil *et al.* 1991; see also Binggeli *et al.* 1988) they will not be considered here.

The luminosity function was calculated for the interval $-21.5 + 5 \log h$ to $-14.0 + 5 \log h$ where the Hubble parameter is defined as $h = H_0/100 \text{ km s}^{-1} \text{ Mpc}^{-1}$, though when generating the samples, we considered a bright cut at -22.5 . A total of 1000 simulations were run using as input parameters $\alpha_{\text{in}} = -0.7, -1.1$ and -1.5 in order to verify the sensitivity of the estimators to the inclination of the faint slope. These values were chosen so as to encompass the range of slopes measured by Marzke & da Costa (1997) when fitting luminosity functions for galaxies in different color ranges. The values of $M^* = -19.1$ and $\phi^* = 0.02$ were used throughout. The simulations were calculated for a survey with a similar geometry to the CfA1, which has a solid angle of 2.66 steradians and an apparent magnitude limit of 14.5, and the number of galaxies varied according to the faint end slope, going from about 1522 to 1734.

The recovered distribution from the Monte Carlo simulations of α and M^* for the case of $\alpha_{\text{in}} = -1.1$ are presented as histograms in Figs. 1 and 2, respectively, while the median values and an estimate of the dispersion obtained in the Monte Carlo simulations are presented in Table 1. In general all methods recover quite well the input values, with the exception of the $1/V_{\text{max}}$ estimators where the recovered values are usually biased either in α or in M^* depending on whether galaxies are binned in redshift or magnitudes. It is immediately apparent that the STY and C^- methods provide the best results. This comes as no surprise, given that these methods use all information available in the sample, since no kind of binning is required. Both recover the input M^* but bias α in different directions. That the STY is somewhat biased had been noted by EEP, but this is within the dispersion value. The C^- presents a slightly larger dispersion than the STY, as can be seen in the figures.

In Table 1 we also list median values for the distribution of α and M^* for two other values of the faint end slope. The corresponding histogram for α is shown in Fig. 3. In these simulations M^* was kept constant. For steeper slopes, most methods tend to be biased in different degrees. The best result is obtained with the C^- , as it recovers the values closest to the input parameters. The STY presents a larger bias than before, and is more than 1σ away from the input value. Because of the correlation between α and M^* , the latter is biased towards fainter (larger) values. This result is somewhat unexpected as a more inclined slope would mean more faint galaxies and therefore a better constraint on the α value (Subbarao *et al.* 1996). For the shallower slope ($\alpha = -0.7$), the results are consistent with those found for $\alpha = -1.1$.

The behavior of the SWML is also worth noting, as it is usually coupled to the STY, in order to provide a visual

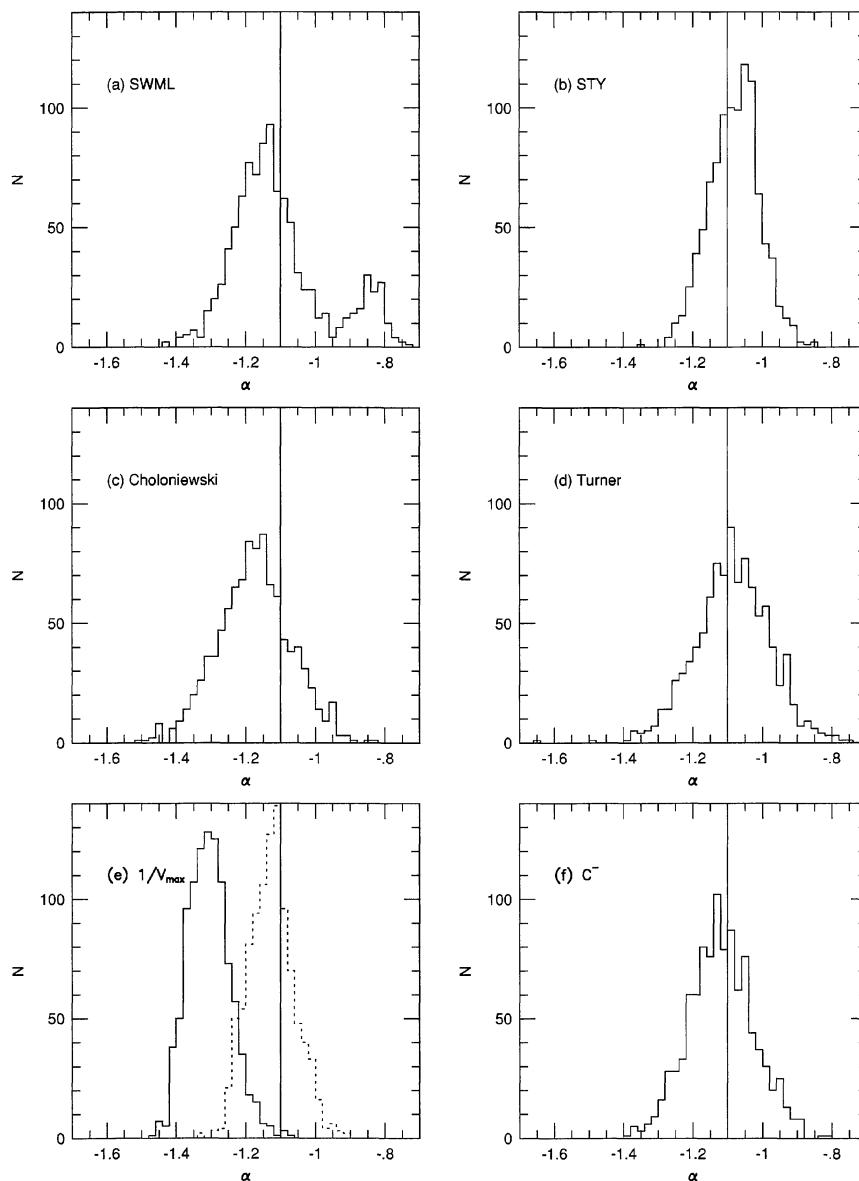


FIG. 1. Histogram showing the distribution of α values, in 0.02 bins. Each panel identifies the method that was used. The vertical line indicates the input value of $\alpha = -1.10$. For the $1/V_{\max}$ estimators, the solid line represents binning in redshift, while the dashed line represents binning in magnitudes.

representation of the latter, and used to estimate the goodness of fit. Here we apply a least-squares fit to the SWML distribution, as a means of estimating the Schechter parameters, and whether the fit parameters present a similar behavior to the STY. An inspection of Table 1 shows that it gives slightly different results, and in Fig. 1 we can see that α recovered from these fits presents a bi-modal distribution. This second peak occurs in samples for which the faint end of the luminosity function is under-represented; the SWML is the only estimator where such a behavior is seen. As may be seen in Fig. 3 this secondary peak gets less defined as one goes to shallower inclinations. All other estimators, in spite of their biases, present a fairly symmetric distribution. A direct comparison between the fit parameters of the SWML

with the STY α and M^* is shown in the panels of Fig. 4. In both figures we plot as dashed lines the value of the input parameters, while the solid line corresponds to equal values on each axis. From the figure one can see that in most cases the SWML fits present more inclined slopes and brighter M^* than the STY maximum likelihood, save those making up the secondary peak seen in Fig. 1. The M^* distribution is presented in Fig. 4(b), and again there is a systematic difference between both measurements.

As described above the C^- method may be considered as a limiting case of the SWML, where only one object per bin is considered. This can be seen by comparing equation (12) (EEP's 2.12 in their paper) with equation (25) in Choloniewski (1987), where the integral luminosity function (obtained via the C^- method) is averaged in small magnitude

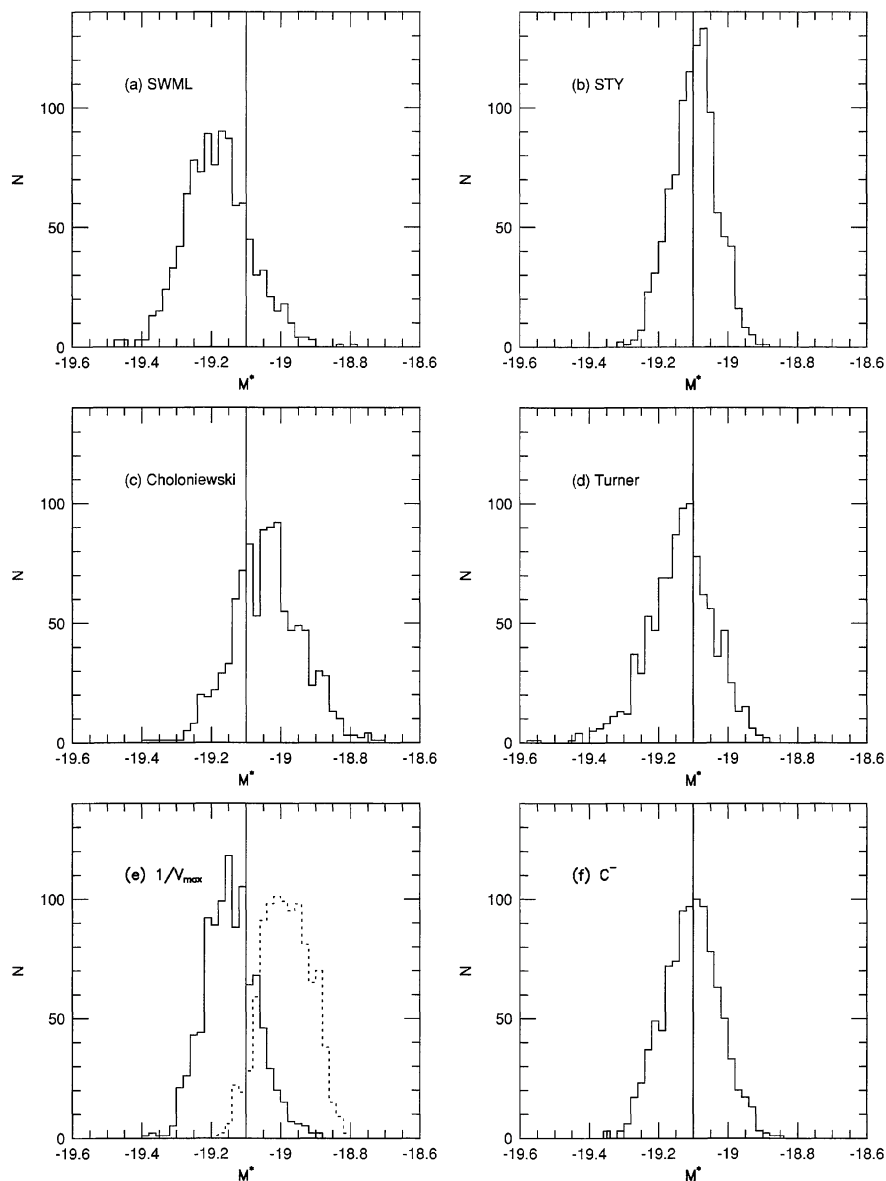


FIG. 2. Histogram of the M^* values recovered from the simulations, for each method as shown in the panel. The vertical line represents the input value of M^* . As in the previous figure, for the $1/V_{\max}$ estimators, the solid line represents binning in redshift, while the dashed line represents binning in magnitudes.

intervals. In practice, both methods present different results, which could be partly explained from the fact that the procedure followed in the computation is different. The SWML is obtained through an iterative procedure where one deter-

mines the differential luminosity function, while the fits based on the C^- method are made using the integrated luminosity function, which, because it is cumulative, is also less sensitive to fluctuations. This is particularly important at the

TABLE 1. Median values for recovered parameters M^* and α , CfA1 like sample.

Method	α	M^*	α	M^*	α	M^*
Input values	-1.10	-19.10	-1.50	-19.10	-0.70	-19.10
SWML	-1.13 ± 0.14	-19.19 ± 0.10	-1.45 ± 0.14	-19.13 ± 0.11	-0.82 ± 0.14	-19.19 ± 0.12
STY	-1.08 ± 0.07	-19.10 ± 0.07	-1.43 ± 0.06	-19.06 ± 0.07	-0.69 ± 0.08	-19.10 ± 0.06
Choloniewski	-1.18 ± 0.11	-19.04 ± 0.10	-1.50 ± 0.08	-19.00 ± 0.11	-0.87 ± 0.15	-19.08 ± 0.12
Turner	-1.09 ± 0.11	-19.13 ± 0.10	-1.51 ± 0.09	-19.16 ± 0.10	-0.66 ± 0.11	-19.11 ± 0.09
$1/V_{\max\Delta z}$	-1.31 ± 0.06	-19.15 ± 0.08	-1.67 ± 0.05	-19.14 ± 0.09	-0.94 ± 0.08	-19.14 ± 0.07
$1/V_{\max\Delta M}$	-1.13 ± 0.07	-18.99 ± 0.07	-1.50 ± 0.06	-18.98 ± 0.07	-0.83 ± 0.11	-19.04 ± 0.09
C^-	-1.12 ± 0.09	-19.10 ± 0.08	-1.51 ± 0.07	-19.11 ± 0.08	-0.72 ± 0.12	-19.10 ± 0.07

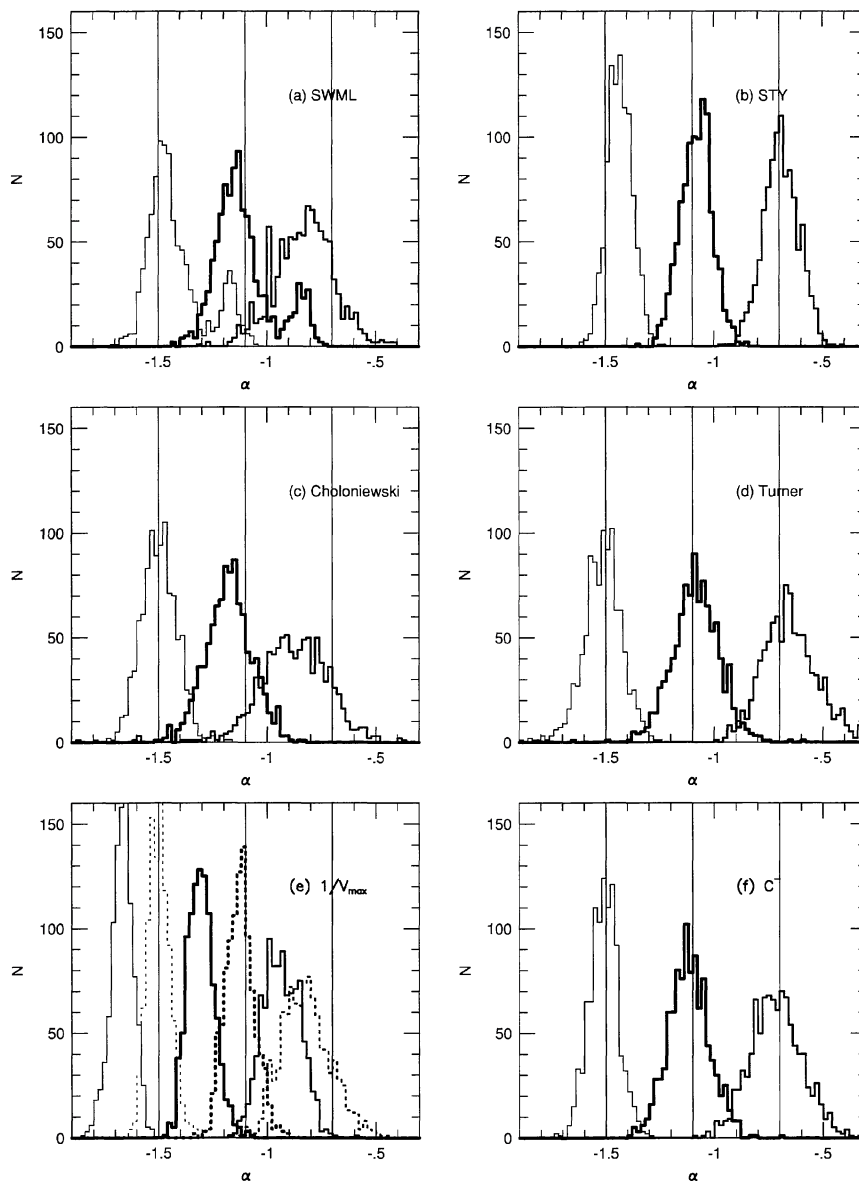


FIG. 3. Histogram comparing the recovered values of α for three different input values represented as vertical lines (-1.5 , -1.1 and -0.7). In the $1/V_{\max}$ method, the solid and dotted lines represent binning galaxies in redshift and magnitudes respectively. For the sake of clarity, the distribution for each input value has been coded with a different line weight.

faint end where the small number of objects can affect the distribution, as in the case of the SWML.

In Table 2 we present values for the luminosity function normalization ϕ^* , obtained in the simulations using as input parameters $\alpha = -1.1$ and $M^* = -19.1$. In all cases, except for ϕ_c^* which is obtained using equation (41), the values were obtained by applying equation (32) to the various mean density estimators. Histograms of the distribution of ϕ^* are presented in Fig. 5, where we show for the $1/V_{\max}$ and Choloniewski methods the normalization obtained by means of a 3-parameter least-squares fits. For the STY, SWML, Turner and C^- methods we present the normalization using the minimum-variance weighting averaged over redshift bins. The use of this weighting on Poisson samples is not entirely correct as the weighting function for a random

sample should be $w(r) = 0$ as $J_3 = 0$. However, the values we recover are not much different from what one measures for the other density estimators. These are presented in Fig. 6, where we show the recovered densities but now only for the STY solutions, mainly because this is the most commonly used method.

Both Figs. 5 and 6 show that the ϕ^* estimators present rather large fluctuations, yet all of these show a trend of underestimating the input value. The minimum-variance estimators present rather similar dispersions, and the estimator which seems to present the best results is that derived from the number counts. It should be noted, that in spite of the observed discrepancies, all estimators agree within $\sim 30\%$ of the input value.

So as to verify the effect that making different absolute

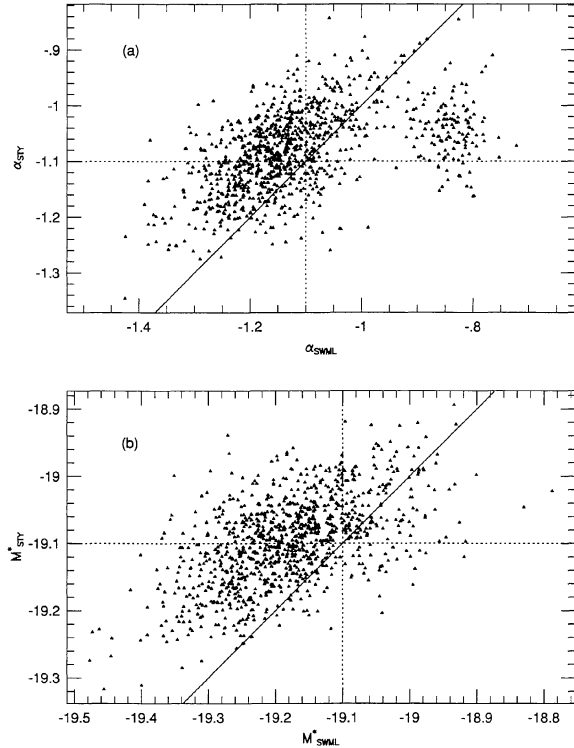


FIG. 4. Comparison between the recovered values for the SWML and STY methods for α (panel a) and M^* (panel b). The solid line represents a line with slope equal to 1, while the dotted lines represent the input values of both parameters.

magnitude cuts could have on the mean density, the luminosity function was calculated for 500 simulations using the cuts in absolute magnitude $-21.5 \leq M \leq -14$ and $-20.0 \leq M \leq -14$. The latter limits are the ones used by Ellis *et al.* (1996) in their study of the evolution of the luminosity function, while the former are close to the limits used by survey efforts such as the SSRS2 (da Costa *et al.* 1997, 1994), the Durham/UKST galaxy Redshift Survey (Ratcliffe *et al.* 1997) and Stromlo-APM (Loveday *et al.* 1992). In general, the ϕ^* estimated for the smaller magnitude interval is higher, but the difference for most estimators is of the order of 5%, demonstrating that the difference in the normalization measured between these works cannot be attributed to the different absolute magnitude ranges.

In order to estimate the sensitivity of the estimators to the number of galaxies in the sample, a series of simulations

emulating the survey of Willmer *et al.* (1994, 1996) was calculated. The latter aimed at measuring redshifts of galaxies down to $b_J=20$ in a $4^\circ \times 0.67^\circ$ slice close to the North Galactic pole. In the simulations the same Schechter parameters as above were used, and no evolution was considered. The results from these simulations are presented in Table 3, while the distribution of recovered α values is shown in Fig. 7. For this smaller sample of galaxies we find that not only do the uncertainties become larger, but for some methods, in particular STY, the biases increase, although this increase also depends on the faint-end slope. For shallower slopes, STY recovers the input parameters very well, while for steeper slopes, it is more than 6σ off, and $M^* 2\sigma$ away from the input values. The C^- is also biased, and this bias tends to get smaller as the slope increases, a behavior opposite to the STY. The bias for M^* is rather constant for the values of α we considered. The SWML is less affected than STY at steeper slopes and overall has a similar behavior to the C^- method, though the magnitude of its bias is always larger. The remaining estimators generally produce better fits when the slopes are steeper, yet more than 1σ away from the input value. The Turner method in particular is extremely sensitive to the number of objects as well as the bin width used in the calculation. Finally, it is interesting to note that for the smaller sample the Choloniewski and $1/V_{\text{max}}$ (in magnitudes) estimators give comparable results.

5. APPLICATION OF THE ESTIMATORS TO THE CfA1 SURVEY

As an illustration of the behavior of the different estimators, and to compare the results here with previous determinations, we have calculated the luminosity function on a sample of real galaxies. For this we chose the CfA1 survey (Huchra *et al.* 1983, 1992), mainly because it has been the sample most extensively studied with different estimators. In Table 4 we compare our measurements with those by other authors, using the sample limits and bin widths (where applicable) as stated in the original papers, which are indicated in the references column. Here it should be noted that all determinations except Choloniewski (1986) used galaxies in both galactic caps. The latter only considered galaxies in the northern galactic cap portion of the CfA1 survey. Whenever possible, we compared the results obtained by other authors for samples without corrections due to the Virgo cluster. The results here show a reasonable agreement with most of the previous determinations, usually within 0.1 both in α and

TABLE 2. Median values for the Schechter function normalization.

Estimator	SWML	STY	Choloniewski	Turner			C^-
				$\times 10^{-2} h^3 \text{Mpc}^{-3}$	$1/V_{\text{max}}(\Delta z)$	$1/V_{\text{max}}(\Delta M)$	
$n(\Delta z)$	1.41 ± 0.33	1.68 ± 0.21	...	1.47 ± 0.26	...	...	1.85 ± 0.25
n	1.30 ± 0.29	1.46 ± 0.19	...	1.41 ± 0.23	...	...	1.81 ± 0.22
$n_1(\Delta z)$	1.69 ± 0.28	1.85 ± 0.21	...	1.77 ± 0.26	...	...	2.10 ± 0.26
$n_3(\Delta z)$	1.72 ± 0.22	1.84 ± 0.18	...	1.79 ± 0.23	...	...	1.70 ± 0.25
n_3	1.87 ± 0.28	2.14 ± 0.21	...	2.03 ± 0.28	...	...	1.47 ± 0.22
n_{EEP}	1.39 ± 0.71	1.80 ± 0.88	...	1.48 ± 0.87	...	...	1.89 ± 1.03
ϕ^*	1.76 ± 0.25	1.97 ± 0.20	...	1.87 ± 0.26	...	...	1.93 ± 0.24
fit	...	...	1.34 ± 0.30	...	1.39 ± 0.19	1.67 ± 0.19	1.80 ± 0.95

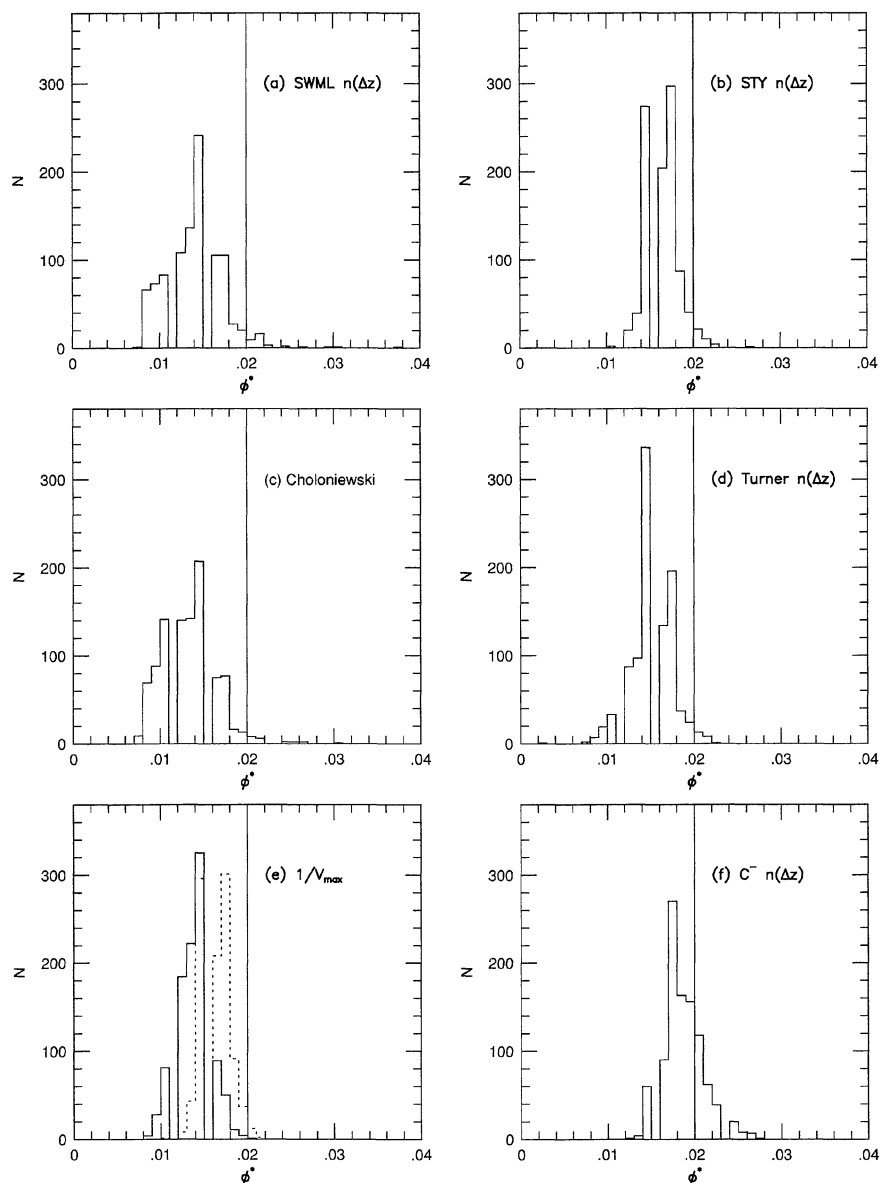


FIG. 5. Histogram comparing a few estimators for the luminosity function normalization. In this figure we show the estimates obtained by making three-parameter least-squares fits to the Choloniewski and $1/V_{\max}$ methods. For the STY, SWML, Turner and C^- methods we use the minimum-variance estimator calculated in redshift bins. The solid vertical line in each panel indicates the input value of ϕ^* of the simulations.

M^* , save in the case of the STY (EEP) method as calculated by EEP and the Turner method using redshift bins of 400 km s^{-1} (DH). The origin of the discrepancy with EEP is unclear, but could be due to the fact that they have taken into account the Malmquist bias introduced by the errors of Zwicky magnitudes. Another source of discrepancy is certainly the catalog itself, as the version that was used here (nz40.dat dating from May 1994, see Huchra 1997) is more recent than any of the papers we compare with.

Another comparison between the different estimators is shown in Tables 5 (for M^* and α) and 6 (for ϕ^*), again using the CfA1 survey, but now considering for all estimators the absolute magnitude range $-21.5 + 5 \log h$ to -14.0

$+5 \log h$, and as above, with no corrections for Virgo, but correcting for the Local Group motion following Yahil *et al.* (1977). The sample contains 2395 galaxies, though in the absolute magnitude interval we considered there are only 2345; all galaxies with zero or negative heliocentric velocities or with corrected velocities less than zero were removed. A plot showing the different fits is presented in Fig. 8, and the inset shows the estimated 1σ error ellipses. In the case of the STY, Turner and C^- methods, the differential luminosity function is not calculated, so no observed distribution is presented. From Table 5 one can see that save for the $1/V_{\max}$, all estimates give fairly consistent results with $\alpha = -1.2$ and $M^* = -19.2$. It is interesting to note that now the STY values

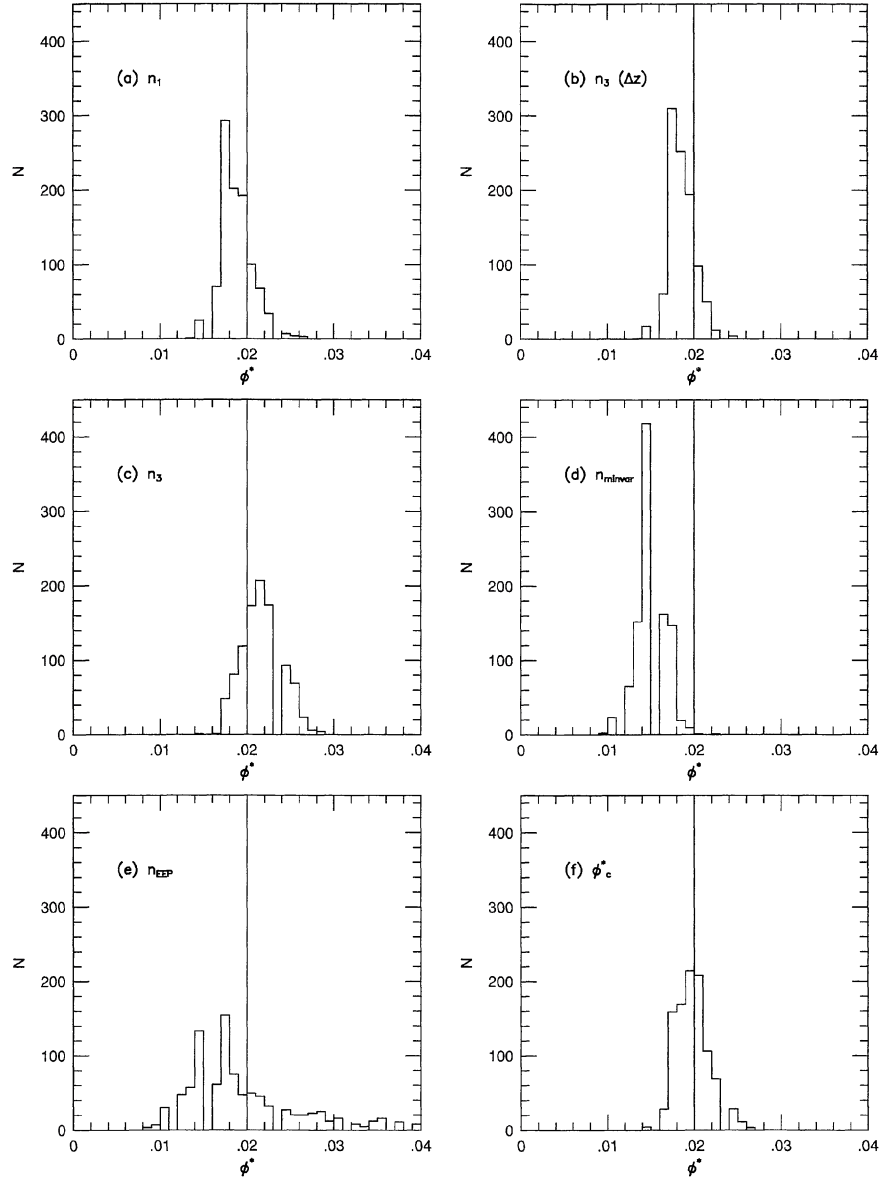


FIG. 6. Histogram comparing six estimators used to calculate ϕ^* but in this case only for the STY method. Panel (a) shows the distribution of ϕ^* obtained using the n_1 estimator of DH in redshift bins, while panel (b) shows the same for the case of n_3 . Panel (c) shows another variant of n_3 , where instead of counting galaxies in redshift bins, one calculates equation (36) using the whole surveyed volume. Panel (d) shows the distribution for the normalization using the minimum variance weighting [equation (34)], but only considering one galaxy per shell. Panel (e) shows the minimum variance calculated using equation (38). Panel (f) shows the normalization using equation (41).

TABLE 3. Median values for the Schechter function parameters, smaller sample.

Method	α	M^*	α	M^*	α	M^*
Input values	-1.10	-19.10	-1.50	-19.10	-0.70	-19.10
SWML	-1.18 ± 0.13	-19.24 ± 0.17	-1.46 ± 0.10	-19.12 ± 0.19	-0.94 ± 0.17	-19.30 ± 0.17
STY	-1.06 ± 0.10	-19.07 ± 0.11	-1.33 ± 0.07	-18.94 ± 0.12	-0.69 ± 0.12	-19.11 ± 0.11
Choloniewski	-1.24 ± 0.14	-19.12 ± 0.19	-1.51 ± 0.11	-19.02 ± 0.19	-0.98 ± 0.19	-19.20 ± 0.19
Turner	-0.82 ± 0.23	-19.07 ± 0.31	-1.25 ± 0.20	-19.12 ± 0.30	-0.78 ± 0.21	-19.20 ± 0.25
$1/V_{\max\Delta z}$	-1.29 ± 0.07	-19.40 ± 0.15	-1.65 ± 0.07	-19.42 ± 0.17	-1.06 ± 0.10	-19.49 ± 0.15
$1/V_{\max\Delta M}$	-1.21 ± 0.08	-19.08 ± 0.12	-1.58 ± 0.07	-19.06 ± 0.14	-0.93 ± 0.14	-19.13 ± 0.13
C^-	-1.16 ± 0.13	-19.15 ± 0.14	-1.53 ± 0.09	-19.14 ± 0.14	-0.79 ± 0.16	-19.15 ± 0.14

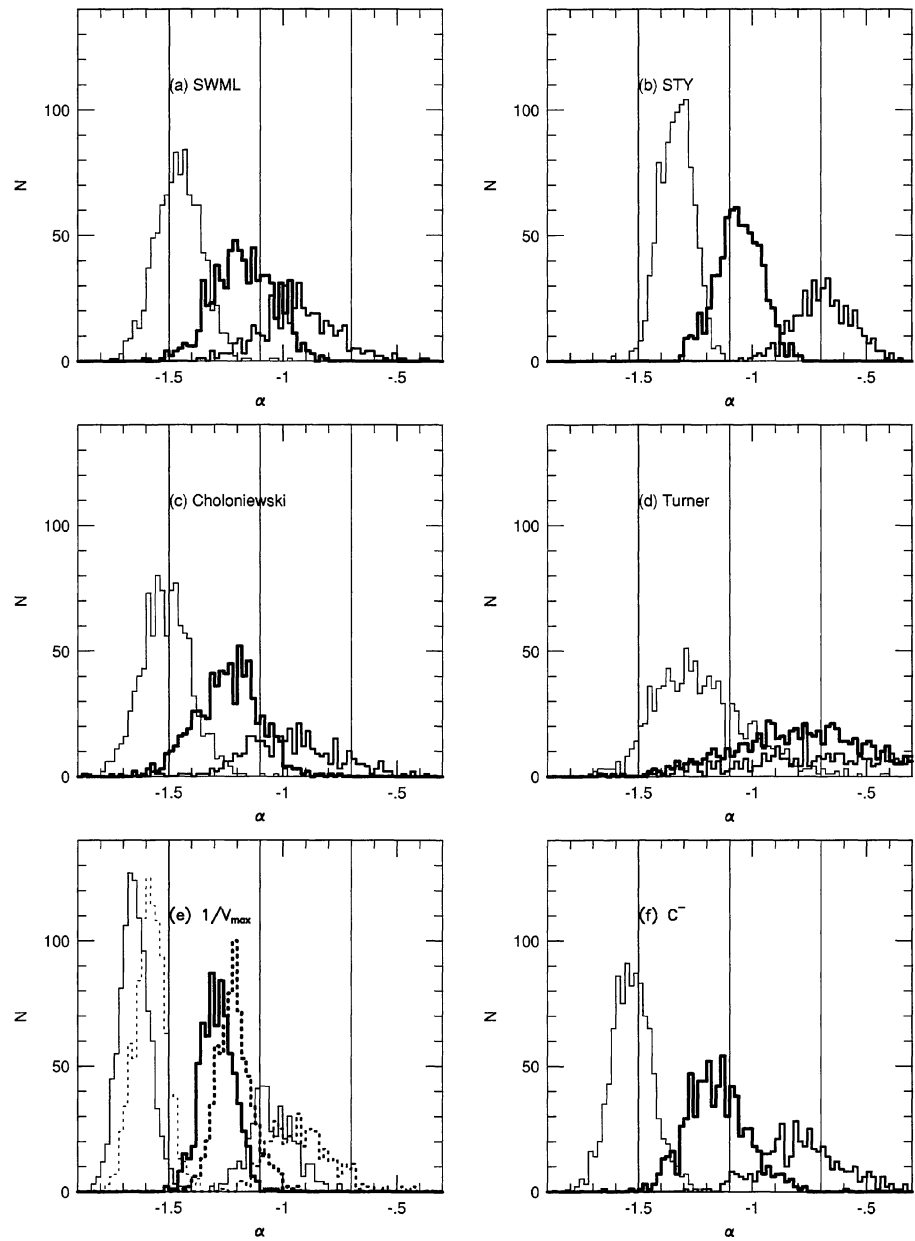


FIG. 7. Histograms showing the recovered α values for samples with ~ 300 galaxies. As can be seen most methods present a very wide distribution for lower values of α . The full and dotted lines for the $1/V_{\max}$ methods represent binning galaxies in equal redshift and magnitude bins. As in Fig. 3, the distribution for each input value has been coded with a different line weight.

TABLE 4. Schechter function parameters for CfA1 survey.

Method	M_{bright}	M_{faint}	α	M^*	Reference	α_{here}	M^*_{here}	Notes
STY	-21.5	-16.0	-1.08	-19.10	EEP	-0.83	-19.00	
$1/V_{\max}$	-21.5	-16.0	-1.5	-19.5	DH	-1.49	-19.28	$\Delta M=0.2$
Turner	-21.5	-16.0	-0.9	-19.2	DH	-1.04	-19.27	$\Delta v=400 \text{ km s}^{-1}$
Turner	-21.0	...	-1.2	-19.2	LGH	-1.10	-19.30	$\Delta v=700 \text{ km s}^{-1}$
Choloniewski	-21.5	-16.0	-1.09	-19.2	Choloniewski (1986)	-1.09	-19.07	$\Delta M=0.25$

TABLE 5. Schechter function parameters for CfA1 survey, using same limits for all methods.

Method	α	M^*	Notes
SWML	-1.20 ± 0.03	-19.30 ± 0.04	$\Delta M = 0.25$
STY	-1.11 ± 0.08	-19.17 ± 0.08	
Choloniewski	-1.18 ± 0.05	-19.26 ± 0.07	$\Delta M = 0.25$
Turner	-1.11 ± 0.06	-19.32 ± 0.05	$\Delta z = 500 \text{ km s}^{-1}$
$1/V_{\text{max}}$	-1.59 ± 0.04	-19.43 ± 0.07	$\Delta M = 0.25$
$1/V_{\text{max}}$	-1.70 ± 0.05	-19.55 ± 0.05	$\Delta z = 500 \text{ km s}^{-1}$
C^-	-1.20 ± 0.01	-19.21 ± 0.01	

are fairly close to the determination of EEP, though the comparison might not be fair, as here there has been no correction for the magnitude errors. The values for ϕ^* present a fairly large dispersion, with a larger concentration of values close to $0.025 h^3 \text{ Mpc}^{-3}$. It is interesting to note that the densities measured using the $1/V_{\text{max}}$ method are lower by a factor of 2 than the estimates using STY.

6. SUMMARY

In this work several estimators for the luminosity function assuming a Schechter form are compared. As would be expected, the methods which use all information available in the sample (such as the STY and C^- methods) give the best results. It is also shown that even for homogeneous samples the $1/V_{\text{max}}$ method using binning gives biased results and in general tends to give higher values for the faint end slope than any of the other estimators. The result of EEP that the STY estimator is slightly biased is confirmed. The STY fit tends to *underestimate* the faint-end slope, giving flatter slopes for a fixed value of M^* . This bias becomes worse for more inclined slopes and smaller samples. A comparison between the least-squares fits to the SWML and the STY maximum likelihood fits suggests that at more inclined values of the faint-end slope, the goodness of fit estimated for STY could be worsened because of the biases of both methods, which tend to grow in opposite directions. This result suggests that for samples with steeper slopes another estimator, such as the C^- , could be more adequate to calculate the non-parametric distribution and fit. We also find that the values for the mean density recovered by most estimators are lower than the input values by factors ranging up to 25%. This implies that the observed discrepancy between samples

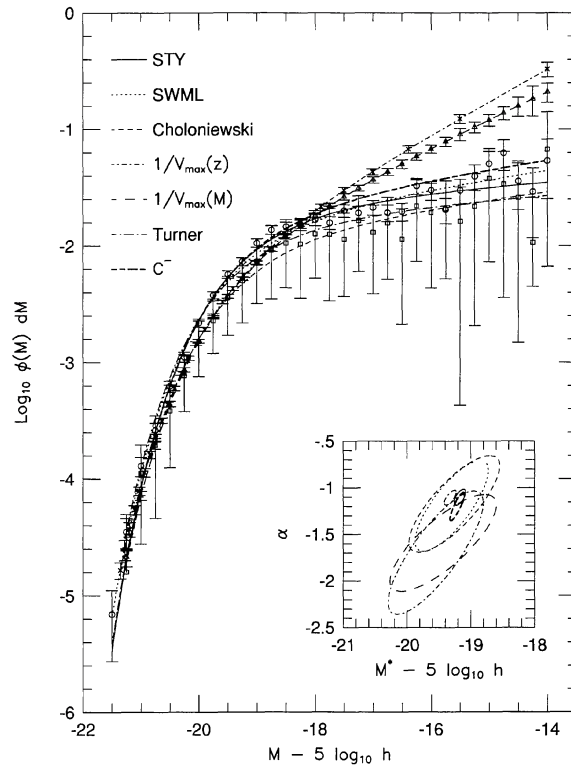


FIG. 8. A plot showing fits obtained for the CfA1 sample using the different estimators. Points represent the observed function, while lines represent the parametric fits. The following estimators are plotted as symbols: SWML (open circles); Choloniewski (open squares); $1/V_{\text{max}}$ in redshift bins (crosses) and magnitude bins (open triangles). The fits for each estimator are coded as noted in the figure. In the case of the STY, Turner and C^- methods, only the fits are presented. The inset shows the error ellipses for the fits.

of local and distant galaxies cannot be attributed to the different estimators used in the analyses carried out by the various groups.

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TABLE 6. Schechter function normalization estimates for CfA1.

Estimator	SWML	STY	Choloniewski	Turner $\times 10^{-2} h^3 \text{ Mpc}^{-3}$	$1/V_{\text{max}}(\Delta z)$	$1/V_{\text{max}}(\Delta M)$	C^-
n (Δz)	1.98	2.60	...	1.74	...	...	2.52
n	1.78	2.32	...	1.58	...	...	2.25
n_1 (Δz)	2.31	3.11	...	2.06	...	...	2.97
n_3 (Δz)	2.18	2.63	...	2.13	...	...	2.46
n_3	2.52	3.04	...	2.48	...	...	2.82
n_{EEP}	1.81	2.88	...	1.47	...	...	2.72
ϕ_c^*	2.76	3.32	...	2.72	...	...	3.09
Fit	...	...	1.36	...	1.00	1.16	2.37

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